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Comparison of the analgesic efficacy of combined superficial and deep serratus anterior plane block versus pectoserratus plane block following modified radical mastectomy surgery

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Abstract

Background Both the pectoserratus plane block and the combined deep and superficial serratus anterior plane block (CSAPB) are recognized as effective regional anesthesia methods for pain control. This research aims to assess the ability of two different nerve blocks to provide analgesia.

Methods The study included 60 patients who underwent modified radical mastectomy (MRM). Participants were allocated into separate groups: Group P (Pectoserratus Plane Block) and Group S (CSAPB). The primary outcome was the tramadol consumption, while the secondary outcome was the assessment of pain levels.

Results The median (Q1, Q3) postoperative 24-hour tramadol consumption was 60 (60,80) mg in Group P and 60 (40,60) mg in Group S. ($p=0.006$) Additionally, higher NRS scores were observed in group P at 2 and 4 h.

Conclusions In this study, we found that the CSAPB and the pectoserratus plane block provided comparable analgesic efficacy in patients undergoing MRM. This approach may serve as an effective alternative to the pectoserratus plane block.

Trial registration number NCT05961735.

Keywords Breast surgery, Multimodal analgesia, Pain, Pectoserratus plane block, Serratus anterior plane block

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Introduction

Among female malignancies, breast cancer continues to be the most commonly diagnosed type [1]. Modified radical mastectomy (MRM) is still widely employed as a principal surgical approach in its management [2]. Postoperative pain leads to reduced patient comfort and prolonged hospital stays. Pain typically originates from trauma to the intercostal nerves along with the subsequent inflammatory processes triggered by this nerve injury. The sensory innervation of the breast is primarily provided by the intercostal nerves originating from thoracic spinal nerves T2 to T6. However, when considering all breast quadrants and the axillary region, additional nerves contribute to the sensory supply. These include the intercostobrachial nerve (T1–T2), the medial and lateral pectoral nerves (C5–C7), and the long thoracic nerve (C5–C7) [3, 4].

The procedure-specific pain management (PROSPECT) guidelines recommend for the incorporation of regional anesthesia techniques, such as paravertebral block (PVB), pectoral nerve blocks and local anesthetic infiltration, into multimodal analgesic strategies to optimize postoperative pain control [5]. Several regional anesthesia methods, including paravertebral block (PVB), pectoserratus plane block, serratus anterior plane block (SAPB), erector spinae plane block and rhomboid intercostal block, have been employed in breast surgery for pain control [6–9]. The SAPB, which can be administered either deep (below) or superficial (above) to serratus anterior muscle, offers effective pain relief [10]. Combined deep and superficial SAP blocks (CSAPB) is recognized as effective regional anesthesia method for pain control [11].

This study hypothesizes that there is no significant difference in the analgesic efficacy between ultrasound-guided pectoserratus plane block and CSAPB for managing postoperative pain in patients undergoing MRM. Specifically, it aims to compare the two techniques in terms of tramadol consumption and postoperative pain scores.

Methods

Trial design

This single-center prospective randomized study complied with the ethical standards of the Helsinki Declaration-2013. Ethics committee approval (No: 034, Date: 05/04/2023) were obtained. (ClinicalTrials.gov: 14.07.2023- NCT05961735) Patients who underwent MRM between August 1, 2023, and November 1, 2023, were included in the study. Written informed consent was obtained from all participants.

Participants and randomization

This study included female patients who underwent MRM. Exclusion criteria included local anesthetic drug allergy, chronic analgesic use, coagulopathy, wound site infection, neuropathy and thoracic deformity. Patients were randomly allocated to groups (group P and group S) using sealed, numbered envelopes. Each patient was assigned a unique identification number, and a blinded anesthesiologist collected postoperative data using these numbers. (Fig. 1)

General anesthesia procedure

An intravenous (IV) line was established, and standard monitoring (electrocardiography, pulse oximetry, noninvasive blood pressure) as well as bispectral index (BIS) monitoring was applied to all patients. Premedication was administered with 2 mg IV midazolam. General anesthesia was induced with 1 mg/kg lidocaine, 2–2.5 mg/kg propofol, 1 mcg/kg fentanyl, and 0.6 mg/kg rocuronium bromide. Following endotracheal intubation, anesthesia was maintained with sevoflurane in a 50% air–oxygen mixture at a fresh gas flow rate of 3 L/min. The concentration of sevoflurane was adjusted to maintain a BIS value between 40 and 60. Intraoperatively, all patients received a remifentanyl infusion at 0.05–0.3 mcg/kg/min IV. Neuromuscular blockade was reversed with 50 mcg/kg IV neostigmine and 10 mcg/kg IV atropine. After extubation, patients were transferred to the postanesthesia care unit.

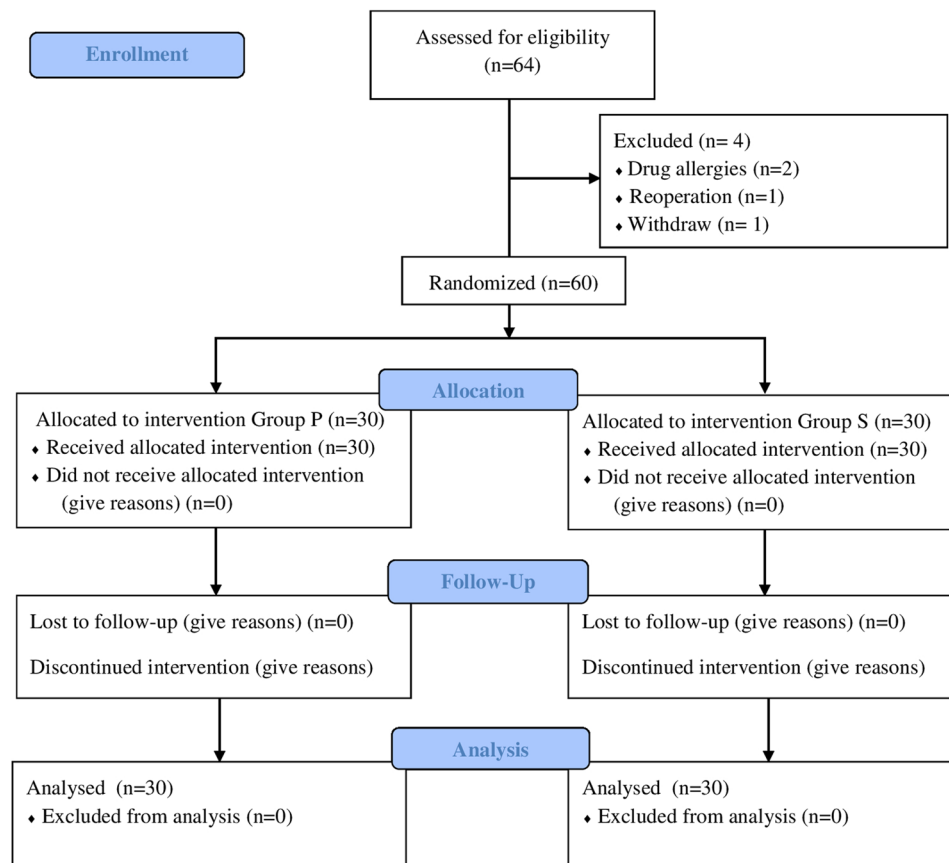
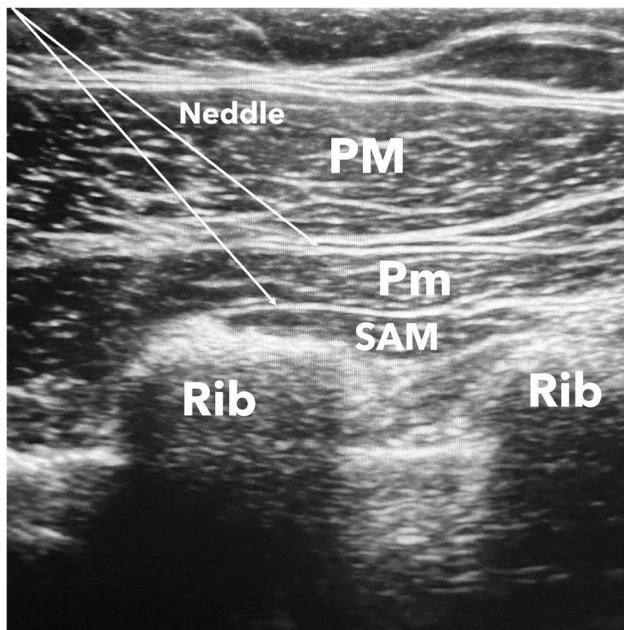
Interventions

Surgery was performed by the same surgical team using the same surgical incision-technique. Pectoserratus plane block or SAPB was administered under ultrasound guidance by the same anesthesiologist. All patients received intravenous ondansetron (4 mg), paracetamol (1 g), and tramadol (100 mg) before the end of the surgery.

Block procedures

During the injection of bupivacaine, intermittent aspiration was performed after every 5 mL to minimize the risk of intravascular administration. Linear ultrasound probe (Sonosite, Bothell, Washington, USA) and 22 G, 50 mm (Echopleks, Vygon) needle were employed for both blocks. Each procedure involved the delivery of 30 mL of 0.25% bupivacaine.

Pectoserratus plane block was performed following the technique outlined in previous studies [7]. The ultrasound probe was positioned along the midclavicular line just below the clavicle and directed inferolaterally to the third rib. After slight medial angulation, the pectoralis major, pectoralis minor, and serratus anterior muscles were clearly visualized. The needle was then inserted into the fascial plane between the pectoralis major and minor

**Fig. 1** CONSORT diagram of the study**Fig. 2** Sonoanatomy of Pectoserratus Plane Block. PM: Pectoralis major muscle, Pm: Pectoralis minor muscle, SAM: Serratus anterior muscle

muscles, and 10 mL of 0.25% bupivacaine was injected. Next, the needle was advanced into the fascial plane between the pectoralis minor and serratus anterior muscles, where 20 mL of 0.25% bupivacaine was administered (Fig. 2).

The CSAPB was performed as described in the literature [10, 11]. Positioned along the mid-axillary line at the fourth rib, the USG probe was used to visualize the latissimus dorsi, teres major, and serratus anterior muscles. Employing an in-plane technique, the needle was advanced toward the fourth rib and placed beneath the serratus anterior muscle, with 15 mL of 0.25% bupivacaine injected. The needle was then retracted 1–2 cm, redirected above the serratus anterior muscle and below the latissimus dorsi muscle, and a second 15 mL of 0.25% bupivacaine was injected. (Fig. 3)

Standard analgesia protocol and outcomes

All patients received intravenous paracetamol (1 g) at 8-hour intervals postoperatively, in combination with patient-controlled analgesia (PCA) containing intravenous tramadol. (20 mg bolus doses, 20-minute lockout period) Postoperative pain was evaluated at five time points: T1 (1st hour), T2 (2nd hour), T4 (4th hour), T12 (12th hour), and T24 (24th hour). In cases where patients

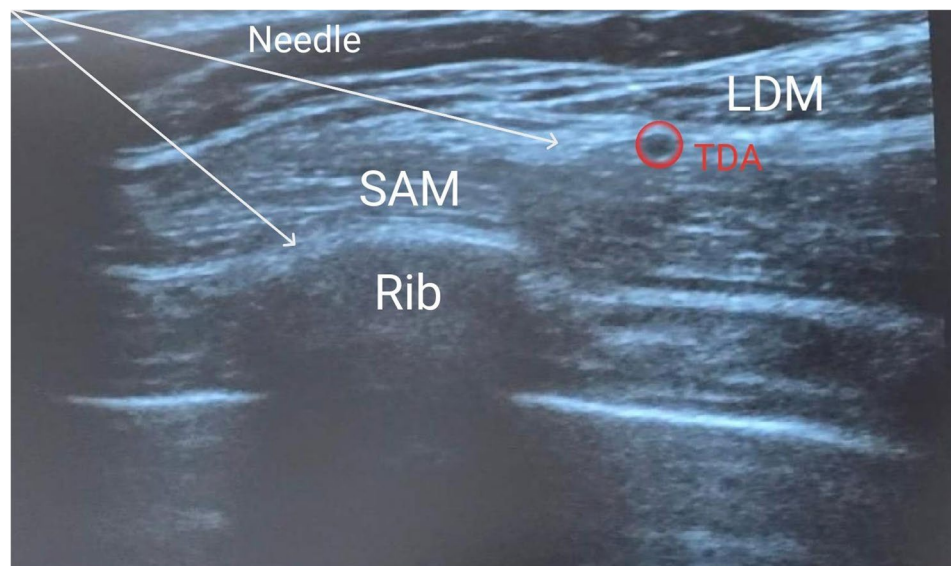


Fig. 3 Sonoanatomy of combined deep and superficial serratus anterior plane block. LDM: Latissimus dorsi muscle, SAM: Serratus anterior muscle, TDA: Thoracodorsal artery

reported NRS score of 4 or above despite the standard analgesic regimen, rescue analgesic (dexketoprofen 50 mg) was administered. The primary outcome was the total tramadol consumption. As a secondary outcome, pain was evaluated using the NRS (numeric rating scale).

Statistical analysis

Statistical analyses were conducted using SPSS. The numerical values were expressed as the mean $\pm$ standard deviation or median (Q1: first quartile- Q3: third quartile). The Kolmogorov-Smirnov test was used to determine data normality. The Chi-square test (for categorical variables), t-test (for continuous variables with normal distribution), and Mann-whitney u tests (for continuous variables with non-normal distribution) were employed in this study. The Bonferroni correction was used to analyze NRS scores, with statistical significance adjusted to $p < 0.01$ owing to measurements from five time points.

Sample size calculation was conducted using G*Power informed by preliminary data on mean tramadol usage. The average tramadol usage was 68 ± 17.88 mg in group P and 56 ± 16.73 mg in group S. (alpha:0.05, power: 80%, effect size: 0.69) Based on this analysis, a minimum of 54 participants was required. Due to possible dropouts, 64 patients were included.

Results

The study initially included 64 female patients. However, 4 participants were excluded: drug allergies (2), reoperation for bleeding (1), and declined to the procedure (1). Ultimately, the study included 60 female patients who underwent MRM. (Fig. 1).

Table 1 Demographic and clinic characteristic

	Group P n = 30	Group S n = 30	p-value
Age (year)	50.96 $\pm$ 8.24	51.80 $\pm$ 10.47	0.733
BMI (kg/m ²)	23.92 $\pm$ 2.07	23.68 $\pm$ 1.30	0.590
ASA score (n)			
I	14	12	0.602
II	16	18	
Surgery time (minute)	96.83 $\pm$ 9.78	94.33 $\pm$ 9.89	0.329

Values are presented as Mean $\pm$ SD and numbers

BMI Body Mass Index, ASA American Society of Anesthesiologists n Number

Table 2 Tramadol consumption and side effects

	Group P n = 30	Group S n = 30	p
Total tramadol consumption (mg)	60 (60,80)	60 (40,60)	0.006
Additional analgesic (n)	3	1	0.301
PONV (n)	3	2	0.640

Values are presented as median (Q1: first quartile- Q3: third quartile) and numbers

N number, PONV postoperative nausea and vomiting

$p < 0.05$ was considered significant

Values in bold type were significant

Demographic and clinical characteristics, including body mass index (BMI), age, and chronic comorbidities, were comparable between the groups. The mean duration of surgery was 96.83 ± 9.78 min in Group P and 94.33 ± 9.89 min in Group S, with no significant difference between the groups ($p = 0.329$) (Table 1).

The median (Q1, Q3) postoperative 24-hour tramadol consumption was 60 (60,80) mg in Group P and 60 (40,60) mg in Group S. ($p = 0.006$) The need for rescue analgesia was similar between the groups ($p = 0.301$) (Table 2).

Table 3 Pain scores

	Group P n = 30	Group S n = 30	p-value
NRS scores at rest			
T1	2 (0, 3)	0 (0, 1)	0.002
T2	2 (0, 3)	0 (0, 1)	0.001
T4	2 (1, 2.25)	1 (0, 1)	< 0.001
T12	2 (1.75, 3)	2 (1, 3)	0.319
T24	1 (1, 2)	1 (0, 2)	0.733
NRS scores at movement			
T1	3 (0.75, 3)	1 (0.75, 2)	0.01
T2	2.5 (2, 3)	1 (1, 2)	< 0.001
T4	2 (2, 3)	1.5 (1, 2)	< 0.001
T12	2 (2, 3)	2 (1, 3)	0.08
T24	1 (1, 2)	1 (0, 2)	0.144

Values are given as median (Q1: first quartile- Q3: third quartile) and numbers

NRS Numerical Rating Scale

$p < 0.01$ was considered significant according to Bonferroni correction

Values in bold type were significant

NRS scores were presented in Table 3. Higher NRS scores (resting, movement) were observed in Group P at 2 and 4 h. No significant differences in NRS scores were observed between the groups at later time points.

Postoperative nausea was observed in three patients in Group P and two patients in Group S ($p = 0.640$).

No complications related to the blocks were noted during either the intraoperative or postoperative periods.

Discussion

In this study, patients who received CSAPB exhibited reduced postoperative tramadol consumption. Additionally, in Group P, resting NRS scores were observed to be higher during the first 4 h, while movement NRS scores were higher at the 2nd and 4th hours. NRS scores were found to be similar between the two groups at subsequent time points.

For effective pain control following breast surgery involving the axilla, it is essential to understand the nerve supply of the chest wall. The lateral pectoral nerve, originating from the C5 to C7 spinal nerves, innervates the pectoralis major muscle, while the medial pectoral nerve, arising from C8 to T1, supplies the pectoralis minor muscle. The thoracic intercostal nerves from T2 to T6 branch laterally and anteriorly to provide sensation to the lateral chest wall and the medial aspect of the breast. Additionally, the long thoracic nerve (C5 to C7) and the thoracodorsal nerve are responsible for innervating the serratus anterior and latissimus dorsi muscles. Pectoserratus plane block targets these nerve groups by depositing local anesthetic between the pectoralis major and minor muscles, as well as above the serratus anterior muscle [7, 10]. CSAPB is performed by injecting local

anesthetic both beneath and above the serratus anterior muscle to achieve nerve blockade [11].

How effective the pectoserratus plane block is in managing pain has been explored in numerous studies within the existing literature. [12, 13]. Alshawadfy et al. reported lower opioid consumption in deep SAPB group when contrasted with pectoserratus plane block [14]. Similarly, in a study comparing thoracic PVB, pectoserratus plane block, and deep SAPB, postoperative opioid requirements were found to be higher in the pectoserratus plane block group [15]. Gupta et al. compared PVB and superficial SAPB and reported similar NRS scores, but noted higher tramadol consumption in SAPB group [16]. In this study, we found that patients who received the CSAPB consumed less tramadol compared to those who received the pectoserratus plane block. Additionally NRS scores were higher in group P during 2nd, and 4th postoperative hours, while scores were comparable between the groups at the 12th and 24th hours. Although lower tramadol consumption was observed in the CSAPB group in our study, the absence of a significant difference in the percentage of tramadol use and the comparable NRS scores beyond the 4th postoperative hour suggest that the two block techniques may provide similar analgesic efficacy. Both the pectoserratus plane block and the SAPB target nerve blockade in the anterior chest wall [7, 10]. The intercostobrachial and long thoracic nerves supply the axillary region with innervation [17]. SAPB, by targeting the intercostal, long thoracic, intercostobrachial, thoracic intercostal, and thoracodorsal nerves, provides analgesic coverage [18]. SAPB has demonstrated greater effectiveness in providing axillary analgesia than the pectoserratus plane block, owing to its capability to block the lateral cutaneous nerve where it traverses the serratus anterior muscle around midaxillary line [10, 14]. We think that the CSAPB may provide effective axillary analgesia due to its capacity to block the lateral cutaneous nerve by delivering local anesthetic both above and below the serratus anterior muscle. This may explain the minimal statistical difference observed in tramadol consumption between the groups.

The efficacy of both deep and superficial SAPB has been investigated in various studies in the literature [19, 20]. In a study comparing deep and superficial SAPB, better analgesic effects were reported with the deep SAPB. However, they did not clearly identify the reasons for the similar analgesic effects of both blocks in the same anatomical areas [21]. A meta-analysis examining deep and superficial SAPB concluded that there was no clear superiority between the two and that both provided similar analgesic effects [22]. Based on the experience of Finnerty et al., it was concluded that deep SAPB is technically easier to perform and provides more easily recognizable

sonoanatomy, especially in obese patients, compared to superficial injection [23]. In the study conducted by Blanco et al. on four volunteer patients, it was reported that superficial SAPB resulted in a longer duration of paresthesia compared to the deep approach [10]. Maranto et al. CSAPB in patients undergoing mastectomy [24]. In two different studies, CSAPB were compared with ESP and paravertebral blocks, and similar analgesic effects were reported for both blocks. It was also noted that multisite injections with local anesthetic distribution to different fascial layers might create a synergistic effect and compensate in case of block failure [11, 25]. Based on the lack of clear superiority between deep and superficial blocks in the literature, and considering the possibility of synergistic effects due to local anesthetic distribution to different fascial layers, we applied CSAPB in this study. Furthermore, since both deep and superficial SAPB are effective in providing pain relief for breast surgery, as well as the combined use of these blocks being reported as an effective pain management strategy in patients undergoing chest wall surgery, we chose to apply both blocks together.

More invasive techniques, such as intercostal nerve blocks and PVB, are associated with a higher risk of complications, including pneumothorax and Horner's syndrome. In comparison, both the pectoroserratus plane block and the CSAPB are considered safer and more reliable techniques. Furthermore, the ability to perform both blocks easily under ultrasound guidance with the patient in the supine position, along with their effectiveness in providing analgesia, are important advantages that enhance their clinical utility [5, 7, 10]. Given its comparable analgesic efficacy and ease of application to the pectoroserratus plane block, which is included in the PROSPECT recommendations, we consider the CSAPB to be a reliable alternative regional anesthesia technique for patients undergoing MRM.

Our study has some limitations. First, as the blocks were performed following the induction of general anesthesia, we could not assess the block levels through dermatomal evaluation. Second, due to the postoperative 24-hour follow-up period, we were unable to evaluate long-term effects, such as chronic pain. Another limitation of our study is the lack of a standardized optimal volume, as different volumes and concentrations of local anesthetics have been used for both blocks in the existing literature. The absence of a control group in the study was also a limitation.

Conclusion

In this study, we found that CSAPB provided effective analgesia in patients undergoing MRM surgery and had a comparable analgesic efficacy to the pectoroserratus plane block, as previously reported in the literature.

This approach may serve as an effective alternative to the pectoroserratus plane block. Furthermore, we recommend conducting multicenter studies to compare the CSAPB with other regional anesthesia techniques.

Authors' contributions

YO: Conceptualization, data curation, formal analysis, investigation, methodology, project administration, resources, supervision, visualization, writing—original draft, writing—review & editing. COY: Conceptualization, data curation, investigation, methodology, visualization, writing—original draft, writing—review & editing. FA: Conceptualization, formal analysis, methodology, project administration, resources, supervision, visualization, writing—review & editing. MZ: Conceptualization, investigation, methodology, project administration, supervision, visualization, writing—review & editing. EA: Conceptualization, methodology, project administration, resources, supervision, visualization, writing—review & editing. MSK: Conceptualization, investigation, project administration, supervision, visualization, writing—review & editing. SA: Conceptualization, formal analysis, methodology, visualization. JE: Conceptualization, methodology, project administration, resources, supervision, writing—review & editing. All authors read and approved the final version of the manuscript.

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Data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

The randomized and prospective trial was conducted after obtaining approval from the Ankara Etlik City Hospital Ethical Committee (Approval No: 034, Date: 05/04/2023). The trial was registered on www.clinicaltrials.gov (<https://clinicaltrials.gov/>) under the identifier NCT05961735 on 14.07.2023.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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